

Jihwan (Ben) Shin

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EDUCATION

ETH Zürich

Sep 2025 - Present

MSc Robotics, Systems and Control

Zürich, CH

- Courses: Probabilistic Artificial Intelligence, Computer Vision, Robot Dynamics, Planning and Decision Making for Autonomous Robots, Computational Models of Motion, 3D Vision, Recursive Estimation
- 3D Vision Project (ongoing): Real-to-Sim Transfer of Bimanual Dexterous Manipulation Tasks in Isaac Sim
- Semester Project (ongoing): Human-Object Interaction Motion Retargeting with Optimisation and RL Supervised by RSL (Robotic Systems Lab) Adria Lopez, Matthias Heyrman, Junzhe He

University of Oxford

Oct 2022 - Jun 2025

BA Engineering Science

Oxford, UK

- Grade: First Class (Final Degree), Distinction (Qualifying)
- Option Courses: Information Engineering Systems, Control Systems, Software Engineering, Biomechanics, Machine Learning
- 3rd Year Group Design Project: Multiagent Autonomous Drone System for Landmine Detection (Winner of the university Gibbs prize for the best project in year group)

United World College of South East Asia

Aug 2019 - May 2022

International Baccalaureate Diploma

Singapore, SG

- Higher Level: Mathematics Analysis & Approaches (7/7), Physics (7/7), Chemistry (7/7)
- Standard Level: English Language & Literature (7/7), Economics (7/7), Chinese Ab Initio (7/7)
- Awarded Physics Academic Award (1st of all Physics students)

EXPERIENCE

NomadZ (Robot Football Team)

Oct 2025 - Present

Motion Control Team Leader (2026-Present), Member (2025-2026)

Zürich, CH

- Developing imitation reinforcement learning algorithm policies for Booster K1 humanoid robot
- Generating motion data from MoCap datasets and in-house video captures

OxRAM Robotics Society (Quadruped Team)

Oct 2022 - Jun 2025

Co-leader (2023-2025), Member (2022-2023)

Oxford, UK

- Managed a team of 7 members to create a 3D-printed 12-DoF quadruped robot
- Designed chassis and leg components of the robot using Fusion 360
- Created the URDF of the robot and implemented the open loop controls on ROS2

Oxford Robotics Institute DRS-C Lab

Jul 2024 - Sep 2024

Summer Research Intern

Oxford, UK

- Supervised by Prof. Ioannis Havoutis and funded by the EPSRC Vacation Internships scheme
- Built a high-level control framework for a 4-DoF 3 Finger Hand robot using ROS2 Control and micro-ROS
- Identified and suggested improvements to the tendon-based hardware design of the robot hand

AWARDS

Casberd Scholarship

2023, 2024

- Awarded by St. John's College (University of Oxford) for excelling in subject examinations

SKILLS

Languages: English (native), Korean (native), Mandarin Chinese (limited working proficiency)

Programming: Python, C++, MATLAB

Frameworks: ROS2, Isaac Sim, Isaac Lab, CasADi

Inspirit AI

Jun 2021

- Participated in a two-week online camp to learn the basics of data analysis and machine learning
- Coordinated to create and present a skin cancer diagnosis model using OpenCV and TensorFlow with 5 international students and an instructor
- Utilized Python libraries (Matplotlib, SciPy, pandas, seaborn) for data analysis and visualization

UKMT Senior Mathematics Challenge 2021

2020,

- Qualified to the British Mathematical Olympiad Round 1 (BMO1)
- Awarded Gold Award with full score received in paper

United World College of South East Asia *International Baccalaureate Diploma* SG

Aug 2019 - May 2022
Singapore,

- Theory of Knowledge (A), Extended Essay - Physics (A)

X-Y Plotter Challenge

Jan 2023 - Mar 2023

- Built a 3D printed pen plotter as part of a programme organized by the Oxford Robotics Institute
- Created a Python program to turn any images into a G-code path that can be drawn by the plotter

OxCrest Generator

([Github](#)) 2023

- Created a Python program to generate a new Oxford college crest design based on a given prompt using a stable diffusion machine learning model as part of a hackathon project

American Mathematics Competition (AMC12)

2020, 2021, 2022

- Qualified to the American Invitational Mathematics Exam (AIME)
- Awarded Certificate of Distinction

PROJECTS

Extended Essay - Double Pendulum and Chaos Motion

([Github](#)) 2021

- Independently researched and wrote an essay on the topic of chaotic motion of a double pendulum
 - Created a Python program to simulate and visualize the effect of changing pendulum parameters
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Lightricity Ltd.

Mar 2024

Research Intern

Oxford, UK

- Participated in a 5-day internship to identify issues in an indoor photovoltaic cell testing equipment through Arduino data collection and circuit analysis

OxRAM Robotics Society (Quadruped Team)

([Github](#)) Oct 2022 - Jun 2025

Co-leader

Oxford, UK

- Managed a team of 7 members to create a 3D-printed 12-DoF quadruped robot
- Designed chassis and leg components of the robot using Fusion 360
- Created the URDF of the robot and implemented the open loop controls on ROS2

Gibbs Prize

2025

- Awarded by University of Oxford for the best 3rd Year Group Design Project in Engineering Science

Technologies: LaTeX, Linux, InDesign, Premiere Pro

